PYL1002: Waves and Oscillations **Tutorial - 1**
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Tuesday, 13/1/2025

PYL1002: Waves and Oscillations – Tutorial 1

Instructions

1. This collaborative tutorial section is designed to allow you to work as a team, collaboratively solving a longer, interesting problem.
2. This exercise has aspects of both analytical and computational elements. Each group should contains at most 4 students. Details of your group along with other Tutorial Instructions can be found on the class Moodle.
3. Each group should work out their solution (following the prompts given) and also implement it in code. This can be done in Excel, or Python, or any other scientific computational resource.
4. Earnest participation in tutorials carry a total of 10% of your final grade in the class. You will not be judged on the accuracy of your final answers, but rather on your participation and effort.
5. Do not try to rush to the final answer. Take your time in the ideation, make sure you understand the logic and rationale of the setup, and how the equations tie to the physical setup at hand.
6. You are encouraged to interact with Prof. Singh and the class TAs as you work on this exercise. No submission is required. After the class, you are encouraged to go and solve/complete the solutions on your own.
7. Be a good team player and participate genuinely. Ask questions, be curious, and have fun.

Laplace's Demon and Not-So-Simple Harmonic Motion

A central goal of classical physics is to be able to predict the state of a system for all of future and all of past. Specifically, if we have a particle in 1D, we would like to determine its trajectory $x(t)$. This is possible because the laws the of physics are *local.* In particular, if you specify the initial position x_0 and initial velocity v_0 of the particle, Newton's 2nd law, $F = m\ddot{x}$, tells us, step by step, how the position and velocity evolve as a function of time. In simple words, if you know where you are, and you know where you're going, $F = ma$ tells where you will be next. And the same process repeats. This is the idea behind *Laplace's Demon* that we briefly discussed in class: an entity with access to every particle's position and velocity at a given time along with enough computational power to use the local laws of physics ($F = ma$) to predict the particle's path for all future and all past. Today, we will try to emulate a simpler version of Laplace's Demon by solving $F = ma$ as algebraic equations, one time step at a time, like clockwork, and apply it to understand some interesting features behind oscillatory motion.

1.1 Taming the Demon with Algebra

We begin by converting the 2nd order differential equation of $F = m\ddot{x}$ into *two, simultaneous first-order* differential equations by using $\dot{x} = v$ and $\dot{v} = F/m$. To convert them into simultaneous algebraic equations, we replace the time derivatives by their difference approximations. For instance, for a function $f(t)$ (which could be the particle's position and/or velocity), we can approximate its derivative as,

$$\left.\frac{df}{dt}\right|_t \approx \frac{f(t + \Delta t) - f(t)}{\Delta t}\,, \tag{1}$$

where Δt is a small time-step over which Laplace's Demon computes the next position and velocity of the particle. This difference approximation has some associated error with it due to Δt being small, but finite.

Given we know the current value of f and df/dt at a given time, this allows us to approximate the value of f at the next time step $t + \Delta t$,

$$f(t + \Delta t) \approx f(t) + \left.\frac{df}{dt}\right|_t \Delta t\,. \tag{2}$$

This is the basic idea behind converting a first-order differential equation into an algebraic equation. Done iteratively and recursively, one can solve for $f(t)$ for many successive time steps.

You would notice that the style of Eq. (1) is that of a forward difference, and similarly, one can also construct a backward difference formula,

$$\left.\frac{df}{dt}\right|_t \approx \frac{f(t) - f(t - \Delta t)}{\Delta t}\,. \tag{3}$$

(a) While we won't prove why (you are welcome to come ask Prof. Singh or the class TAs why), there is a way to combine Eqs. (1) and (3) to get a much better approximation for the first derivative. Show that by combining Eqs. (1) and (3), one can write the derivative as,

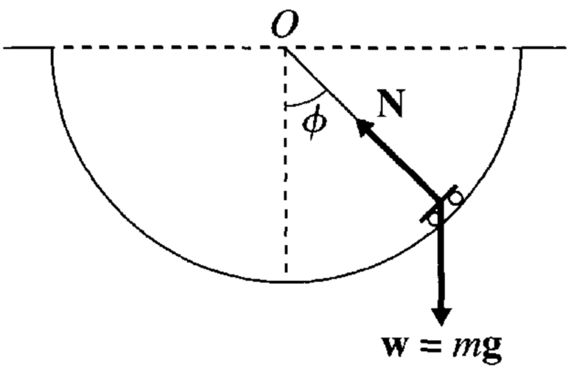
$$\left.\frac{df}{dt}\right|_t \approx \frac{f(t + \Delta t) - f(t - \Delta t)}{2\Delta t}\,. \tag{4}$$

This is known as the "central difference" implementation of the first derivative and has much lower error than either the forward or backward difference formulas.

(b) Next, use this central difference implementation to write down $F = m\ddot{x}$ as a set of two, simultaneous algebraic equations, one which solves for $x(t + \Delta t)$ and one for $v(t + \Delta t)$ based on values at t. Keep this iterative, recursive algorithm ready since we will soon implement it numerically.

1.2 Laplace's Demon at the Skatepark

While visiting IIT Delhi, Laplace's Demon visits the Sports Complex and finds a semi-circular skatepark as shown schematically in the figure below. In class, we wrote equations of motion for the pendulum which is very similar in setup to this skatepark (only the tension is now replaced by the normal force).



(c) Instead of breaking the force vector along horizontal and vertical components, this time write the equation of motion (that is $F = ma$) for the skater in a direction perpendicular and normal to the track. You should prove the equation of motion of the angle ϕ to be,

$$\ddot{\phi} = -\frac{g}{R}\sin\phi\,, \tag{5}$$

where R is the radius of the skate track, and g is the acceleration due to gravity.

This differential equation has no solution in terms of elementary functions. While the small angle approximation to this differential equation gives us the standard harmonic oscillatory functions, we are not going to restrict ourselves to the small angle approximation here. After all, you cannot curb Laplace's Demon to small angles.

(d) Write down central difference algebraic equations for $\phi(t)$ and $\dot{\phi}(t)$ using the implementation from section 1.1. What was x is now ϕ and what was $\dot{x}$ is now $\dot{\phi}$, the logic remains unchanged.

You need do this next part in a "coding" environment. You can use what you prefer, like Python, if you're comfortable with it. Whatever code you write must be your own and not through any smart AI-style assistance.
You can also use ***Google Sheets*** *to do so. To help you get started with Google Sheets, you may check this template: https://tinyurl.com/laplacedemon, where we have implemented the solution of a standard first-order differential equation $\dot{x} = 2x$ using the forward-difference scheme based on Eq. (1) with initial conditions at $t = 0$ of $x(t = 0) = 1$. The solution is an exponential which you can verify. Feel free to adapt this for the second-order differential equation we are interested in using a central difference implementation.*

(e) Now, write a computational program to implement recursively the algebraic equations you have developed to solve *numerically* for $\phi(t)$ and $\dot{\phi}(t)$ for $N = 750$ time steps, $t_{j+1} = t_j + \Delta t$ with $j = 0, 1, 2, \cdots (N - 1)$. Take the skater to be launched from rest at an initial angle of $\phi_0 = \pi/4$ rad. You can take $R = 2$ meters for the radius. You should use the central difference implementation of first derivatives. To do so, take the first two time steps to have the same initial values of $\phi_0 = \pi/4$ and $\dot{\phi}_0 = 0$, and then proceed using the central difference from there (This is a standard trick to bootstrap second-order difference schemes). You can take $\Delta t = 0.05$ sec as a representative of a small time-step. You are encouraged to experiment with smaller or larger Δt and observe what breaks.

(f) Using your solution, plot in your program ϕ as a function of time for at least three oscillation cycles and mark your axes properly.

(g) Run this for various initial conditions. See how the time period varies as a function of this amplitude. Try to verify from your numerical results that for small angle oscillations, the time period is $\tau = 2\pi\sqrt{R/g}$ independent of the amplitude ϕ_0.

In class today, we got to the following point in finding the time period of oscillation for the pendulum,

$$T = 4\sqrt{\frac{l}{2g}} \int_0^{\phi_0} \frac{d\phi}{\sqrt{\cos\phi - \cos\phi_0}} . \tag{6}$$

We then did a Taylor series expansion for $\cos\phi - \cos\phi_0$ and got,

$$\cos\phi - \cos\phi_0 \approx \frac{(\phi_0^2 - \phi^2)}{2}\left[1 - \frac{(\phi_0^2 + \phi^2)}{12}\right] . \tag{7}$$

Continuing from where we left off in class. We can now invert under the square root, we get the following approximation (we used another Binomial approximation for the term in the square bracket) for the integrand of Eq. (6),

$$\frac{1}{\sqrt{\cos\phi - \cos\phi_0}} \approx \sqrt{\frac{2}{\phi_0^2 - \phi^2}}\left[1 + \frac{(\phi_0^2 + \phi^2)}{24}\right] \tag{8}$$

(h) Now, use ChatGPT or Gemini or something similar (you can use it since you are being given express permission to do so, the remainder of the parts should not use AI) to evaluate the integral of Eq. (6) with the approximation of Eq. (8). You should be able to show that the time period of the non-linear pendulum to leading order depends quadratically on the amplitude,

$$T(\phi_0) \approx \tau\left(1 + \frac{\phi_0^2}{16}\right) \tag{9}$$

1.3 More Interesting Things to Explore...Time Permitting

(i) If you still have time remaining (kudos for getting here!), you are encouraged to solve for the standard projectile path of a particle in a 2D plane under Earth's gravity using your Laplace demon algorithm.

That's all for now, folks! More next time.